Kai Qin TAN

Kuala Lumpur, MY | +6017-3791218 | kqtan99@gmail.com | Portfolio | LinkedIn Profile | GitHub Profile

Data Scientist with a strong foundation in Banking, Actuarial Science and experience delivering measurable business impact through **advanced analytics**, **machine learning**, and **AI solutions** in **regulated financial environments**. Currently supporting **business banking** in a **digital bank**, driving revenue growth, strategic decisioning, and operational efficiency through data science initiatives. Proven in **designing** and **developing** scalable, backend ML solutions end-to-end, and partnering with product, engineering, and business teams to **translate insights into action**. Technically proficient in **Python**, **ML/AI**, **Snowflake**, and cloud-based GenAI tools, with a strong focus on production-ready models and sustainable analytics.

EXPERIENCE

Data Science, Specialist | GX Bank Berhad | Petaling Jaya, MY

- Specialised in **Business Banking Data Science and Analytics**, supporting both GX and GXS Bank (SG-based).
- Conducted **root-cause analysis** on the MSME onboarding funnel, delivering data-driven insights to improve conversion and customer experience.
- Led **Business Banking High Depositor Analysis** for GXS, building client profiles using a **LightGBM model** to identify high-value customers with 90% AUC.
- Delivered end-to-end **campaign analytics** for Operating Accounts and Business Loans, analysing customer acquisition, drawdown behaviour, and campaign effectiveness to support strategic decision-making.

Data Scientist | Sun Life Malaysia Assurance | Kuala Lumpur, MY
January 2024 – Present

- **Developed** machine learning models using **XGBoost** and **neural networks** to predict client payment defaults, achieving a **70% accuracy rate**, supporting client campaign initiatives.
- **Reengineered** a legacy **SAS e-miner Cross-sell model** in Python with **Scikit-Learn**, fully documented to meet internal compliance and audit standards, improving maintainability.
- **Built** an OCR-driven Python application to automate data extraction and analysis, **unlocking RM800k/month in new business opportunities**, a 30% extra sales opportunity.
- Led the implementation of **Anaconda Python** across analytics teams, eliminating ~**RM300k in annual SAS licensing costs**, and introducing **GitLab**, **Apache Airflow**, and **Linux** to the team.
- **Collaborated** with vendors on a GenAI proof-of-concept using Amazon Bedrock and Business Q, showcasing the potential of **LLMs trained on internal knowledge bases** to automate client queries and insights.

Intern, Strategic Analytics | Sun Life Malaysia Assurance | Kuala Lumpur, MY
October 2022 – January 2023

- Developed the Campaign Management module for an internal-use drag-and-drop platform using Flask framework and SQLite server.
- Led the backend team in A/B testing software logic using the tree-node concept.
- Constructed a regression duration predictor model for the data extraction process with 90 percent accuracy.

PROJECT

Client Next Best Action (NBA), Sun Life Malaysia Project

- Designed and developed a **client journey framework** tailored to distinct personas and backgrounds, supporting the company's Next Best Action strategy for **personalised client engagement**.
- Built a machine learning model to predict **clients at risk of missing upcoming payments**, aimed at improving customer retention, increasing policy persistency, and **reducing conservation costs**.
- Achieved **70% accuracy** in identifying high-risk clients; model implementation is projected to **reduce conservation costs by 30%** and enable targeted, cost-effective interventions.

Formula One Race Winner Prediction, Personal Project

- Developed a machine learning model to classify F1 drivers into podium, points, or non-scorer categories with **90% accuracy**, using historical performance data and driver-specific attributes.
- Engineered a **custom scoring algorithm** to quantify a driver's track-specific performance potential, incorporating contextual and personal data.
- Built and deployed a full-stack web application using **React and Flask**, integrating race calendars and driver profiles via external APIs, hosted as a PaaS solution.
- Demonstrated **end-to-end ownership**—from open-source data collection and feature engineering to model deployment and frontend integration.

EDUCATION

Bachelor of Science in Actuarial Science with Distinction | Universiti Tunku Abdul Rahman (UTAR)
October 2020 - October 2023

- **CGPA**: 3.97/4.00
- **Awards**: MDT Innovations Top Graduate, Jobstreet Best Graduate (LKCFES), Little Rain Trust Best Graduate 2024, 11 times President List

- **SKILLS:** Python, Machine Learning, Natural Language Processing, Analytics & Insights, SQL, R, Flask, Linux RedHat, Solution Design
- **TOOLS:** Tensorflow, Scikit-Learn, PyTorch, Anaconda, Snowflake, SAS, Amazon Bedrock, GitLab CI/CD, Excel VBA, Docker, Apache Airflow
- **LANGUAGES**: English (Full Working Proficiency), Malay (Working Proficiency), Mandarin (Native), Japanese (JLPT N4)

CREDENTIALS

- **AWS re/Start program Graduate** and a **AWS Certified Cloud Practitioner**.
- **Microsoft Data Science certificate** program graduate.
- Represented the AS course and gave a talk to the existing foundation students on the course and its structure.

REFERENCES

Dr. Wong Voon Hee, Assistant Professor, UTAR
HoD, Department of Mathematical and Actuarial Sciences
wongvh@utar.edu.my | +6012-3552644